

Without ICEPOT

You're absolutely right — "Test the cart functionality" is far too vague. A strong **shopping cart test scenario** should cover the full lifecycle of cart interactions, pricing logic, and edge cases. Here's how we can strengthen it:

✅ Strong Scenario: Shopping Cart

- **Add to cart** Verify adding single/multiple items, different product types, and quantity increments.
- **Remove from cart** Ensure items can be deleted individually or all at once, and cart updates correctly.
- **Update quantity** Validate increasing/decreasing item counts, handling minimum (1) and maximum stock limits.
- **Price calculation** Confirm subtotal, tax, shipping, and grand total are accurate after every cart change.
- **Coupon/discount handling** Test valid/invalid coupons, percentage vs fixed discounts, and stacking rules.
- **Out-of-stock behavior** Ensure unavailable items cannot be added, and cart updates if stock changes after addition.
- **Persistence** Check cart retention across sessions, devices, and after login/logout.
- **Performance** Validate cart updates under load, response times, and concurrency handling.

This way, the scenario isn't just "cart works" — it's a **comprehensive validation of cart reliability, accuracy, and user experience**.

Would you like me to expand this into a **full test suite** (manual + automation candidates) like we did for search, or keep it at the **scenario level** for now?

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Here's how I'll approach **Phase 1** of your request as a Senior QA Test Architect:

🔍 Clarification Questions

To design a complete enterprise-level test suite for the **Shopping Cart**, I need clarity on the following:

1. Application Domain

- Is the cart part of an e-commerce, grocery, food delivery, pharmacy, electronics, fashion, or B2B marketplace?
- Domain impacts inventory rules, tax logic, shipping, and cart persistence.

2. Supported Features

- Which of these are in scope: save for later, move to wishlist, coupons/promo codes, gift cards, multi-currency, cart merge, persistent cart, guest cart?
- If unsupported, I'll exclude them.

3. Quantity Rules

- What are the **minimum and maximum quantity limits** per product?
- Are fractional quantities allowed (e.g., 1.5 kg in grocery)?
- Should setting quantity to **zero** remove the product?

4. Inventory Management

- How is stock validated? Real-time sync with backend?
- Should users see **low-stock warnings**?
- What happens if a product becomes **out-of-stock** after being added?

5. Pricing & Discounts

- Are there **rounding rules** (e.g., 2 decimal places)?
- Do we support **stackable coupons** or only one at a time?
- Is there a **free shipping threshold**?

6. Tax & Shipping

- Are taxes **inclusive or exclusive**?
- Do shipping charges vary by **location, weight, or product type**?

7. Persistence & Sessions

- Should carts persist across **logout/login**?
- Should carts merge when a **guest logs in**?
- What is the **session timeout** behavior?

8. Error Handling

- What are the expected error messages for invalid inputs (e.g., invalid coupon, out-of-stock)?

- Should errors be **blocking** or allow partial checkout?

9. Performance & Scalability

- What is the **maximum cart size** (number of products)?
- Expected **concurrent users**?
- Any SLAs for **response time**?

10. Security & Compliance

- Should we validate against SQL injection, XSS, and authorization?
- Any compliance requirements (e.g., PCI DSS for payments)?

11. Accessibility

- Should the cart meet WCAG 2.1 standards?
- Do we need screen reader support, keyboard navigation, ARIA labels?

12. API Validation

- Is the cart exposed via **REST/GraphQL APIs**?
- Should API-level tests be included?

13. Cross-Platform Scope

- Should we validate on mobile, desktop, and cross-browser?
- Any specific browsers/devices to prioritize?

Missing Requirements

- **Domain confirmation** (e-commerce, grocery, etc.)
- **Feature scope** (wishlist, coupons, gift cards, etc.)
- **Quantity rules** (min/max, fractional, zero behavior)
- **Inventory sync behavior** (real-time, batch, warnings)
- **Pricing rules** (rounding, stacking discounts, free shipping)
- **Tax/shipping logic** (inclusive/exclusive, calculation rules)
- **Persistence/session rules** (guest vs logged-in, merge, timeout)
- **Error handling expectations** (blocking vs non-blocking)
- **Performance SLAs** (cart size, concurrency, response time)
- **Security compliance** (PCI DSS, OWASP validation)

- **Accessibility standards** (WCAG, ARIA)
- **API exposure** (REST/GraphQL)
- **Cross-platform scope** (mobile, browsers, devices)

👉 Once you confirm these details, I'll generate the **Phase 2 full enterprise-level test suite** with structured test cases across functional, negative, BVA, EP, exploratory, regression, smoke, security, performance, accessibility, and automation feasibility.

Would you like to start by clarifying the **application domain** first, or should we go feature-by-feature?